

TESTING WITH JEST AND JQUERY

Ensuring DOM Reliability in Modern Web Applications

THE FOUNDATIONS

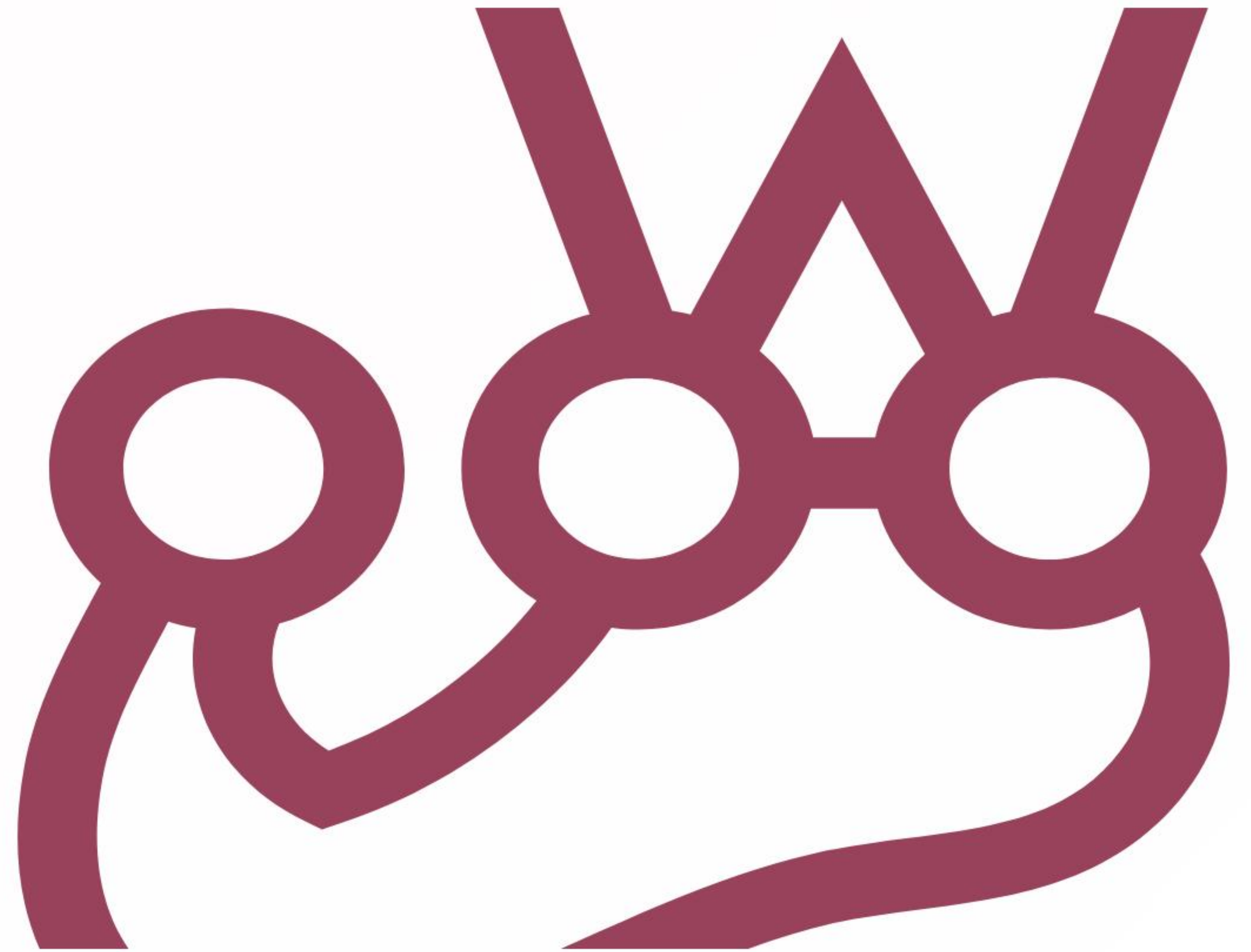
Combining the simplicity of Jest with the utility of jQuery.

INTRODUCTION TO JEST

A Delightful Framework

Jest is a JavaScript testing framework designed for simplicity and speed. It provides built-in support for assertions, mocking, and code coverage.

It integrates seamlessly with JSDOM, a browser environment simulator that allows you to run tests in a Node.js environment without a real browser.



WHY TEST JQUERY?

Legacy & Reliability

While modern frameworks like React exist, many mission-critical systems still rely on jQuery for DOM manipulation and event binding.

Testing these interactions ensures that UI updates happen as expected and that legacy scripts don't break during refactoring or environment updates.



ENVIRONMENT SETUP



Dependencies

Install `jest-environment-jsdom` and `jquery` to ensure your tests can access the DOM simulation.



Configuration

Set your test environment to `jsdom` in your `package.json` or `jest.config.js` file to enable global window objects.



Global Access

Optionally bind `jQuery` to the global object so your scripts can use the familiar `$` variable during tests.

THE AAA PATTERN

Arrange & Act

Set up your innerHTML with the required DOM structure. Then, execute the script or trigger the jQuery action you want to test.

Assert

Use Jest's `expect()` syntax to verify the DOM state. Check for text changes, CSS classes, or visibility attributes.

TESTING DOM SELECTION

Verify that elements are selected and modified correctly by your scripts. This includes checking visibility and existence.

- ❑ Check for specific class names (`.hasClass`)
- ❑ Validate text content (`.text()`)
- ❑ Confirm element visibility (`.is(':visible')`)
- ❑ Monitor attribute changes (`.attr()`)



EVENT LISTENERS

100%

INTERACTION COVERAGE

Triggering Events

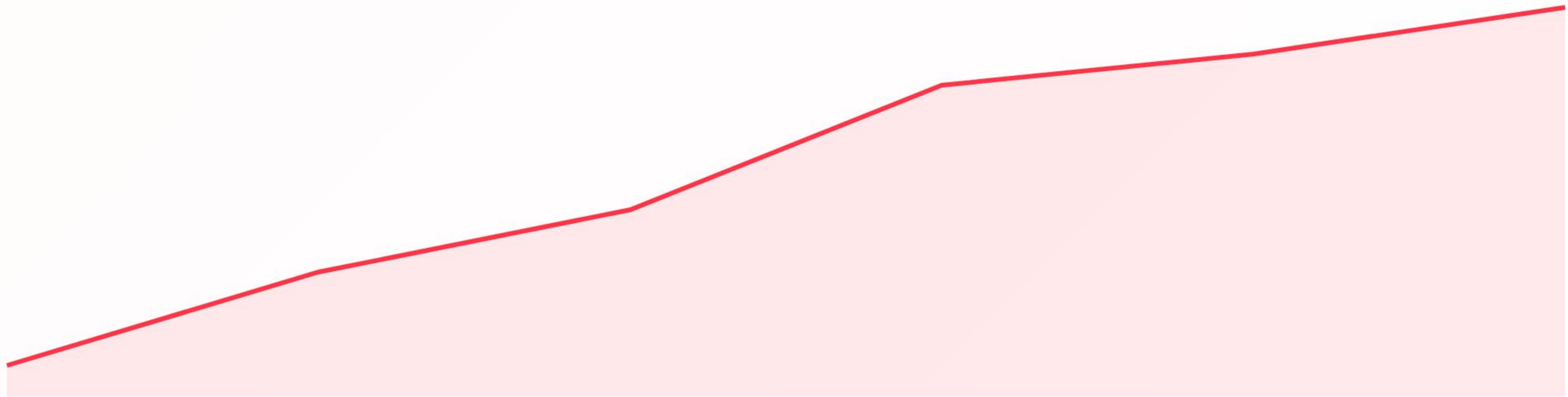
Jest allows you to simulate user behavior by triggering events directly on jQuery objects. This tests your click handlers and input listeners effectively.

Common actions include `.click()`, `.trigger('submit')`, or simulating key presses to ensure your logic responds to user input.

MOCKING AJAX REQUESTS

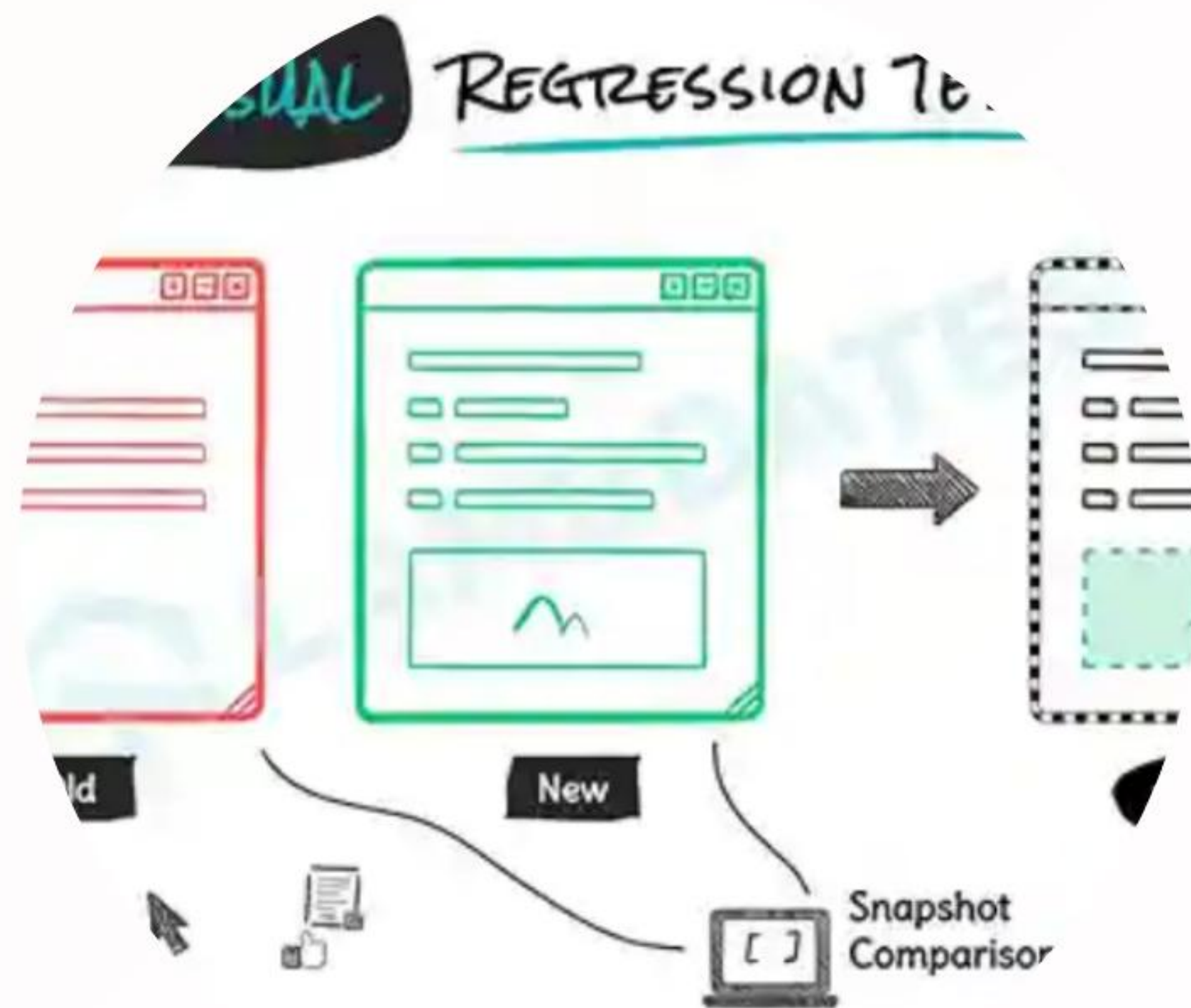
Scenario	Testing Method	Outcome
Data Fetching	jest.fn() implementation	Success data returned
Server Errors	mockImplementation(error)	Error handler triggered
Loading States	Simulated delay	Spinner visibility verified

COVERAGE PROGRESSION



The graph illustrates the growth of test coverage over a 6-month period, demonstrating the transition from manual testing to an automated Jest-jQuery suite.

SNAPSHOT TESTING



UI Regression Detection

Jest snapshots allow you to save the entire DOM structure rendered by your jQuery scripts. If a future update changes the markup unintentionally, the test fails.

This is extremely powerful for ensuring that your dynamic DOM modifications remain consistent across different versions of your application.

QUESTIONS?

Thank you for your time.

www.jestjs.io | www.jquery.com

IMAGE SOURCES



<https://icon.icepanel.io/Technology/svg/Jest.svg>

Source: techicons.dev



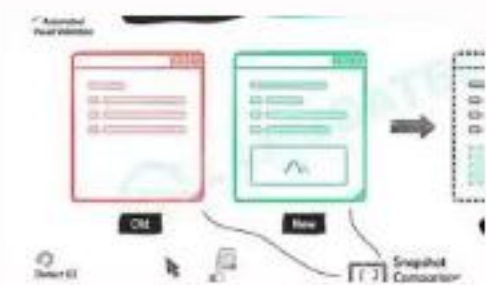
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